Stanisław Szufa

University of Geneva, Switzerland · stanislaw.szufa@unige.ch · +41 76 472 16 09

Prof. Nate Foster
Chair of the Faculty Recruiting Committee
School of Computer and Communication Sciences
EPFL, Lausanne, Switzerland

Dear Professor Foster and members of the Committee,

I am delighted to apply for the position of Tenure Track Assistant Professor advertised in your call for Faculty Positions in Computer and Communication Sciences. I work in computational social choice, the study of voting and collective decisions with the tools of computer science. I have proved complexity results, such as the hardness of deciding how similar two elections are, and built the data and software needed to test voting rules empirically. I am a postdoctoral researcher at the University of Geneva, hosted by Roland Bouffanais and Didier Wernli. Before that, I was a CNRS postdoc at LAMSADE, Paris Dauphine-PSL, with Jérôme Lang, and I received my PhD from AGH University of Kraków in 2024, advised by Piotr Faliszewski.

My best-known contribution is the *map of elections*. I proposed a distance that compares elections regardless of how their candidates and voters are labeled, which made it possible to place real and synthetic elections in a common space and to see, for example, that real elections occupy only a small part of it. The framework (*Artificial Intelligence*, 2025) now extends to approval elections, stable matching, tournaments, and fair division. I also founded Pabulib, the largest open library of participatory budgeting data, now cited in more than 100 papers. The team I belong to also convinced Polish cities, including Wieliczka and Świecie, to adopt the Method of Equal Shares, a proportional voting rule. In total, I have published 5 journal articles and 28 full conference papers, mostly at IJCAI, AAMAS, AAAI, and NeurIPS. As a PhD student, I was the principal investigator of a National Science Centre grant (NCN Preludium); my thesis received the Polish Prime Minister's Award, and I also received a START fellowship and the AAMAS 2023 Best Student Paper Award.

At EPFL, I will bring these tools to collective decisions in which opinions are written in text and aggregated by AI, for example to measure which opinions an AI-written summary of a public consultation leaves out. In Geneva, I already work with votes over texts. I lead a study of all 101,070 abstentions in the UN General Assembly, with each resolution's subject labelled by a language model, and I am a coauthor of a simulation of democratic deliberation among LLM agents (Findings of EMNLP 2026). My agenda would add computational social choice to IC, and it relates to Bryan Ford's work on e-voting and proof of personhood and to Antoine Bosselut's work on NLP and open multilingual language models such as Apertus. I would also like to continue this work in Switzerland, where I live, and where citizens vote on public questions several times a year.

At Jagiellonian University, I taught three undergraduate courses. I have also co-presented the tutorial *Experiments in Social Choice* at IJCAI 2023, AAMAS 2023, and AAAI 2024, and co-supervised more than ten BSc and MSc students, whose work led to papers at AAAI 2025 and ECAI 2025. At EPFL, I can teach core bachelor courses such as programming, discrete mathematics, and algorithms. In my own courses, students will apply each concept to real data, and I would like to develop a master course on algorithms for collective decisions and a seminar on AI and democracy.

Please find enclosed my CV, publication list, research statement, and teaching statement; my references are listed in the application. Thank you for considering my application. I look forward to hearing from you.

Yours sincerely,

Stanisław Szufa